Math 10C — Unit 7: Assignments

The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

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1. Each equation below represents a relation. Without sketching the graph, sort the letters into the LINEAR row or the NON-LINEAR row.

(a) $y = -6x + 11$ (b) $y = \dfrac{7}{x}$ (c) $y = 8x$ (d) $y = 2 - 5x^2$ (e) $y = 5^x$ (f) $y = \frac{2}{3}x - 1$ (g) $y = 4 + x^3$ (h) $y = 10 - \frac{1}{2}x$

LINEAR: _____

NON-LINEAR: _____

2. State the slope and y-intercept of the graph of each linear equation.

(a) $y = 9x - 1$

(b) $y = \dfrac{5}{8}x + 4$

(c) $y = 10 - \dfrac{2}{3}x$

(d) $5y = 15x - 20$

(e) $y = px + q$

3. Write the equation of the line with the given slope and y-intercept.

(a) slope $= 6$, y-intercept $= -7$

(b) slope $= \frac{1}{3}$, y-intercept $= \frac{3}{4}$

(c) slope $= -5$, y-intercept $= 0$

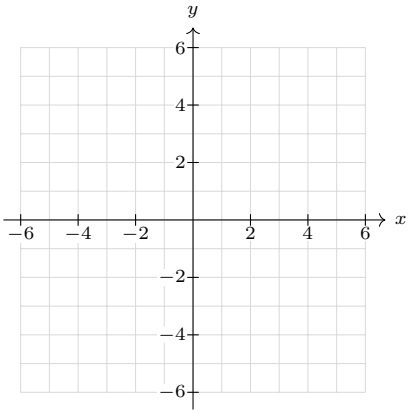
(d) slope $= k$, y-intercept $= n$

4. For each line, state the slope and the y-intercept, then graph the equation by hand.

(a) $y = \dfrac{1}{2}x - 3$

slope = _____

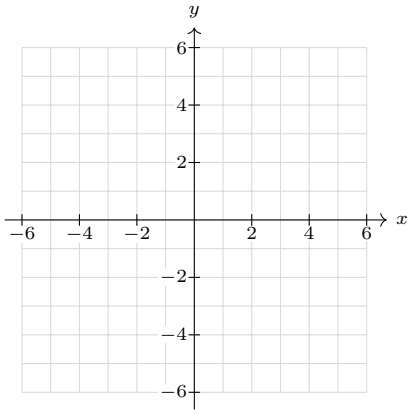
y-int = _____



(b) $y = -x + 4$

slope = _____

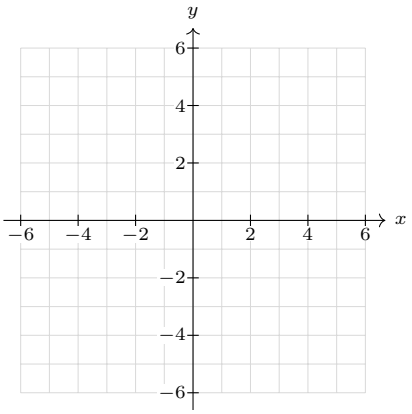
y-int = _____



(c) $y = -\frac{5}{6}x$

slope = _____

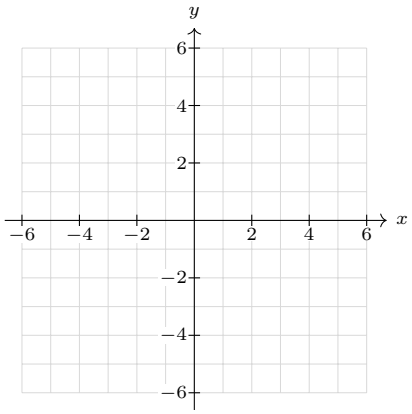
y-int = _____



(d) $y = -2$

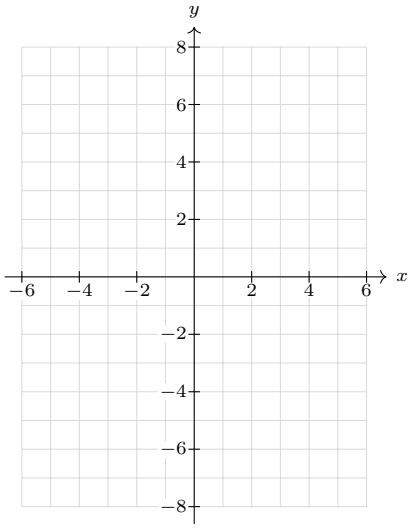
slope = _____

y-int = _____

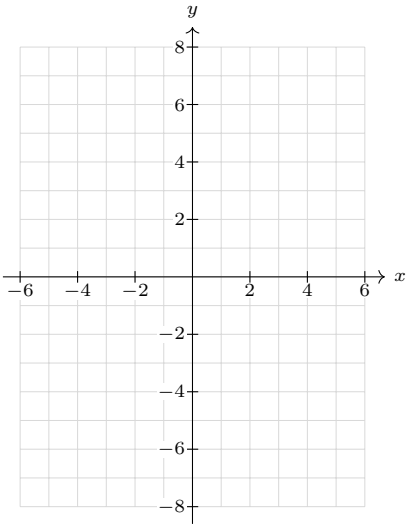


5. For each line, state the y-intercept, determine the x-intercept algebraically, and graph the equation by hand.

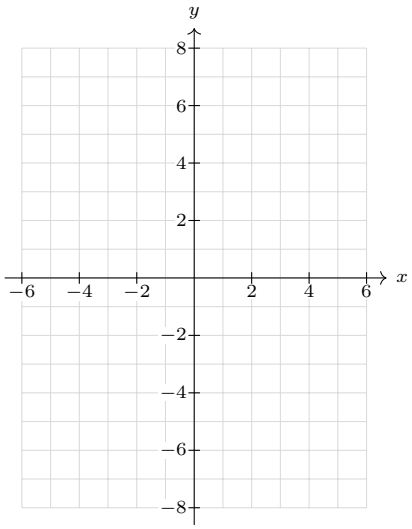
(a) $y = 4x + 8$



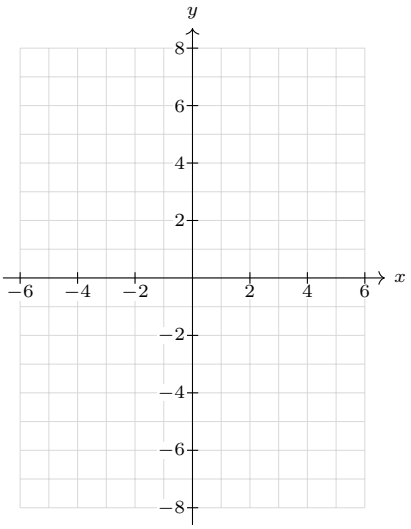
(b) $y = -x - 6$



(c) $y = \frac{4}{3}x - 8$



(d) $y = -\frac{3}{4}x + 3$



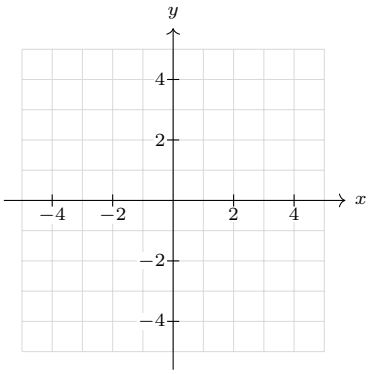
6. Explain why the linear equation $y = 8x$ can be graphed using the method in Question 4 but not by the method in Question 5.

7. Consider the graph of the function with equation $y = -x$.

(a) State the values of m and b.

(b) Determine the x- and y-intercepts.

(c) Sketch the graph.



(d) Determine the domain and range of the function.

(e) How does the graph of $y = -x$ compare with the graph of $y = x$?

8. For each linear equation, determine the slope, the y-intercept, and the x-intercept to the nearest hundredth.

(i) $y = 9x - 14$ (ii) $y = -\frac{29}{2}x - 40$ (iii) $y = 50 - \frac{7}{4}x$

9. *(Multiple Choice)* Which of the following does *not* represent the equation of a straight line?

A. $y = 4x$ B. $y = 15 - 2x$ C. $y = \frac{5}{x}$

D. All of the above represent the equation of a straight line.

10. *(Multiple Choice)* Which of the following statements is *false* for the line $y = -\frac{1}{3}x + 2$?

A. The graph falls from left to right.

B. The x-intercept is 6.

C. The graph passes through the point $(9, -1)$.

D. The line is perpendicular to the line $y = -3x + 5$.

11. *(Multiple Choice)* Which of the following statements is *true* for the line $4y = \frac{1}{3}x + 8$?

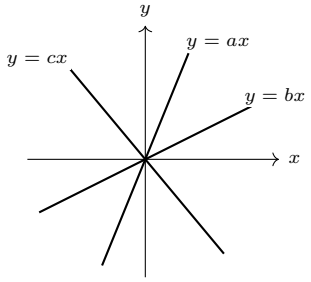
A. The x-intercept is 24.

B. The y-intercept is 8.

C. The slope is $\frac{1}{12}$.

D. The graph passes through the point $(-8, 3)$.

12. *(Multiple Choice)* The lines $y = ax$, $y = bx$ and $y = cx$ are shown below. Which of the following statements is true?



A. $a < b < c$ B. $b < a < c$

C. $c < a < b$ D. $c < b < a$

Use the following information to answer Questions 13 and 14. Consider the line with equation $y = 5x + 4$. The line intersects the x-axis at P and the y-axis at Q. Triangle POQ is formed, where O is the origin.

13. *(Numerical Response)* The area of $\triangle POQ$, in square units, to the nearest tenth, is _____.

Record your answer left to right. | | | | |

14. *(Numerical Response)* To the nearest tenth, the perimeter of $\triangle POQ$ is _____.

Record your answer left to right. | | | | |

Lesson 2 Assignment *Show your work.*

1. Write the equation of each line.

(a) with slope 6 and y-intercept -2

(b) with a y-intercept of 5 and a slope of $-\frac{3}{7}$

(c) passing through the origin with a slope of $-\frac{4}{9}$

(d) with y-intercept 8 and parallel to $y = x$

(e) with a y-intercept of -11 and perpendicular to $y = -\frac{5}{6}x + 2$

(f) with the same y-intercept as $y = x - 9$ and parallel to $y = \frac{3}{8}x + 1$

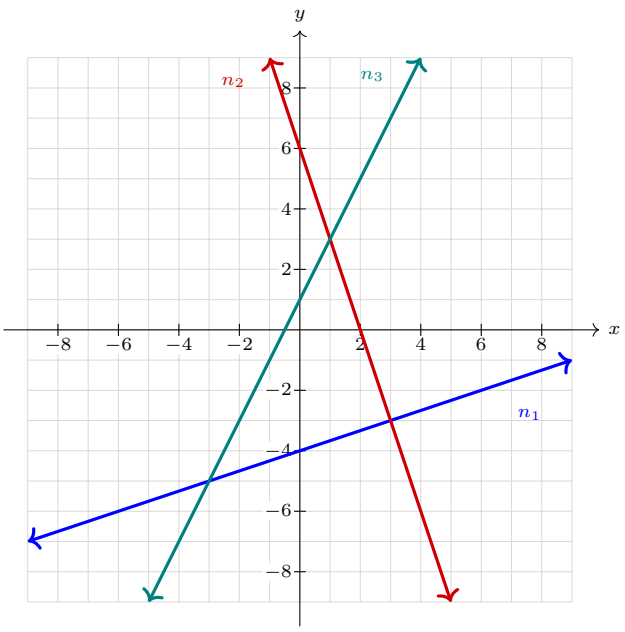
(g) through the point $(0, -2)$ and perpendicular to $y = 7x + 4$

(h) through the point $(0, 10)$ and parallel to $y = \frac{1}{12}x - 30$

(i) with the same y-intercept as $y = 4x + 11$ and perpendicular to $y = \frac{9}{2}x - 5$

(j) with the same y-intercept as $y = px + q$ and perpendicular to $y = rx + s$

2. Each of the lines on the grid passes through points with integer coordinates. Determine the equation of each line.



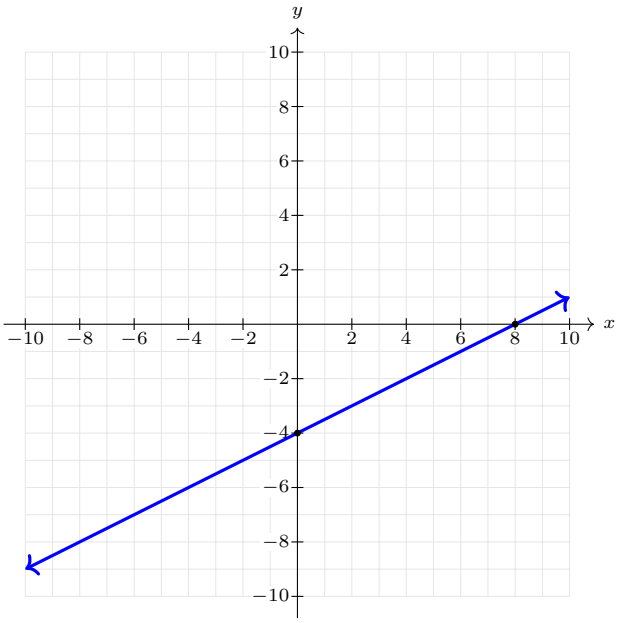
n_1:

n_2:

n_3:

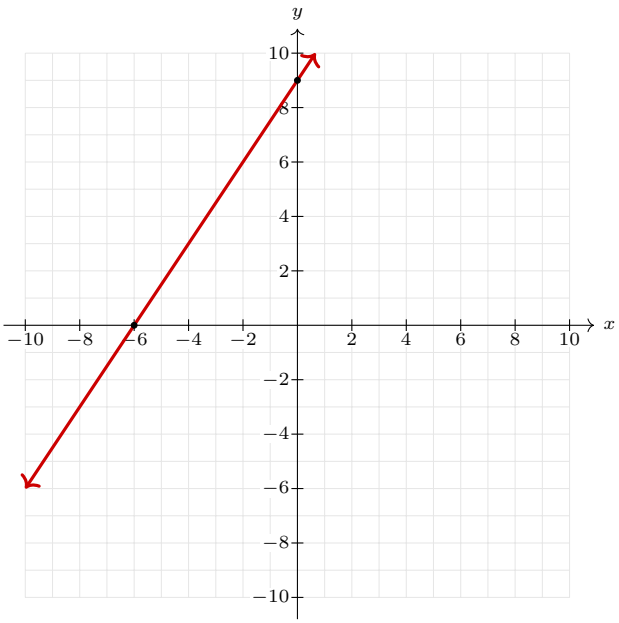
3. Each graph below shows a line whose x- and y-intercepts are both integers. Complete the table for each graph.

(a)



x-intercept	
y-intercept	
slope	
equation	

(b)



x-intercept	
y-intercept	
slope	
equation	

4. Determine the equation of the line which passes through the point $(0, -14)$ and is parallel to the line which passes through $(2, 9)$ and $(5, -3)$.

5. Determine the equation of the line which passes through the point $(0, 6)$ and is perpendicular to the line which passes through $(4, -9)$ and $(10, -11)$.

6. State the equation of each line.

(a) through the point $(-8, 5)$ and parallel to the y-axis

(b) through the point $(-8, 5)$ and parallel to the x-axis

(c) through the point $(2, -6)$ and parallel to the x-axis

(d) through the point (a, b) and parallel to the y-axis

7. Consider the linear function with equation $y = 5$.

(a) State the values of m and b.

(b) State the x- and y-intercepts of the graph.

(c) Determine the domain and range of the function.

(d) Determine the point of intersection of this line with the line $y = 5x - 10$.

8. Consider the relation with equation $x = -7$.
(a) State the x- and y-intercepts of the graph.

(b) Explain why this relation is not a function.

(c) Determine the domain and range of the relation.

(d) State the equation of the line parallel to the y-axis that passes through $(3, -7)$.

9. Write the equation of each line.
(a) parallel to the x-axis, through $(7, -2)$

(b) parallel to the y-axis, through $(7, -2)$

(c) perpendicular to the x-axis, through $(-3, 9)$

(d) perpendicular to the y-axis, through $(-3, 9)$

(e) the x-axis

(f) the y-axis

(g) through $(2, -5)$, perpendicular to the line $y = 3$

(h) through $(-4, 1)$, parallel to the line $x = 6$

(i) through $(5, 7)$, perpendicular to the line $x = -1$

10. *(Multiple Choice)* A line is parallel to the y-axis and passes through the point $(9, -4)$. The equation of the line is
A. $x = 9$ B. $x = -4$ C. $y = 9$ D. $y = -4$

11. *(Multiple Choice)* A line is parallel to the x-axis and passes through the point $(-12, 7)$. The equation of the line is
A. $x = -12$ B. $x = 7$ C. $y = -12$ D. $y = 7$

12. *(Multiple Choice)* The line through the origin, perpendicular to the line with equation $y = \frac{5}{4}x$, has equation

A. $y = \frac{5}{4}x$ B. $y = \frac{4}{5}x$

C. $y = -\frac{5}{4}x$ D. $y = -\frac{4}{5}x$

13. *(Numerical Response)* Consider the line which is perpendicular to $y = \frac{1}{5}x + 7$ and has the same y-intercept as $y = 4x - 13$. If the equation of this line is written in the form $y = mx + b$, then the exact value of $m - b$ is _____.

Record your answer left to right. | | | | |

14. *(Numerical Response)* Two perpendicular lines intersect on the y-axis. One line has equation $y = 5x + 2$. If the equation of the other line is $y = mx + b$, then the exact value of $m + b$ is _____.

Record your answer left to right. | | | | |

Lesson 3 Assignment *Show your work.*

1. Convert the following equations from slope–intercept form ($y = mx + b$) to general form ($Ax + By + C = 0$), where A, B and C are integers, $A \geq 0$, and A, B and C have no common factor.

(a) $y = 6x - 5$ (b) $y = -3x + 8$ (c) $y = px + q$

(d) $y = -\frac{5}{6}x + 3$ (e) $y = \frac{3}{4}x + \frac{1}{8}$ (f) $y = \frac{7}{2}x - \frac{1}{3}$

2. Determine the slope and y-intercept of the graph of each of the following lines.

(a) $x + y - 15 = 0$ (b) $4x - 3y + 24 = 0$ (c) $9x - 4y - 8 = 0$

(d) $5x + 10y - 15 = 0$ (e) $6y = 3x + 24$ (f) $5x + 4y = 20$

3. Determine the slope, y-intercept and x-intercept of the graph of each of the following lines.

(a) $3x + y - 9 = 0$ (b) $7x - 4y + 28 = 0$ (c) $6x - 7y - 4 = 0$

4. Write the equation, in general form, of a line parallel to $5x - 2y + 10 = 0$ and with the same y-intercept as $12x - 2y - 8 = 0$.

5. Write the equation, in general form, of a line perpendicular to $4x - 5y + 10 = 0$ and with the same y-intercept as $6x - y + 24 = 0$.

6. Consider the lines $2x - 3y + 6 = 0$ and $8x + ky - 12 = 0$.

(a) Determine the value of k if the lines have the same slope.

(b) Determine the value of k if the lines have the same y-intercept.

7. Consider the lines $4x - 6y - 24 = 0$ and $ax + 3y - 9 = 0$.

(a) Determine the value of a if the lines have the same slope.

(b) Determine the value of a if the lines have the same x-intercept.

8. Match each equation on the left with the correct characteristic of its graph on the right. Each characteristic may be used once, more than once, or not at all.

Equation

(i) $4x - 6y + 9 = 0$ _____

(ii) $3x - 7y = 0$ _____

(iii) $x + 5y + 10 = 0$ _____

(iv) $x - 6y + 18 = 0$ _____

(v) $3x - y - 6 = 0$ _____

Characteristic

A. Slope $= -\frac{1}{5}$ B. y-intercept $= -5$

C. Passes through $(7, 3)$ D. Slope $= 0$

E. y-intercept $= \frac{3}{2}$ F. y-intercept $= 3$

G. x-intercept $= 2$

9. *(Multiple Choice)* The slope of the line with equation $8x + 5y - 3 = 0$ is

A. $-\frac{8}{5}$ B. $-\frac{5}{8}$ C. $\frac{8}{5}$ D. $\frac{3}{5}$

10. *(Multiple Choice)* Which line has a y-intercept of -4?

A. $x + 4y - 4 = 0$ B. $x + 3y + 3 = 0$

C. $x - 2y - 8 = 0$ D. $2y = 5x - 4$

11. *(Multiple Choice)* The slope of a line perpendicular to the line $x + 4y + 12 = 0$ is

A. -12 B. $-\frac{1}{4}$ C. $\frac{1}{4}$ D. 4

12. *(Multiple Choice)* The line $3y + 4x + 12 = 0$ intersects the y-axis at P. The slope of the line joining P to $Q(9, -6)$ is

A. $-\frac{2}{9}$ B. $\frac{2}{9}$ C. $-\frac{4}{3}$ D. $-\frac{9}{2}$

13. *(Multiple Choice)* The lines with equations $ay = 6x + 15$ and $y = 3x - 8$ are perpendicular. The value of a is

A. 2 B. $-\frac{1}{18}$ C. 6 D. -18

Use the following information to answer Question 14.

Consider the following statements about all the lines in the form $kx + 6y - 18 = 0$, where $k \in R$.

Statement 1: The lines have the same slope.

Statement 2: The lines have the same y-intercept.

Statement 3: The lines have the same x-intercept.

14. *(Multiple Choice)* Which of the above statement(s) is/are true?

A. 1, 2, and 3 B. 1 only C. 2 only D. 3 only

15. *(Multiple Choice)* Line L has equation $4x - 5y + 26 = 0$. A is the point $(-4, 2)$, B is $(6, 4)$, and C is $(-9, -2)$. Which of these points lie on line L?

A. A only B. A and B only
C. A and C only D. B and C only

16. *(Numerical Response)* Given that the line joining the points $(1, 1)$ and $(11, -p)$, where $p \in W$, is perpendicular to the line $5x - 3y - 9 = 0$, then the value of p is _____.

Record your answer left to right. | | | | |

Lesson 4 Assignment *Show your work.*

1. State the equation, in slope–point form, of the line through the given point and with the given slope.
(a) $(7, 2)$, slope 3 (b) $(5, -4)$, slope -2 (c) $(-6, 8)$, slope 5

(d) $(0, -5)$, slope $\frac{1}{3}$ (e) $(-10, 0)$, slope $\frac{3}{5}$ (f) $\left(-2, -\frac{3}{2}\right)$, slope $\frac{4}{7}$

2. Write each equation in slope–intercept form, $y = mx + b$.
(a) $y - 4 = 6(x - 3)$ (b) $y + 2 = -5(x - 1)$ (c) $y - 6 = -9(x + 5)$ (d) $y + 8 = 3(x + 15)$

3. Find the equation, in slope–intercept form, of the line through the given point and with the given slope.
(a) $(3, 7)$, slope 4 (b) $(4, -3)$, slope 5 (c) $(0, 6)$, slope -3

(d) $(-8, 3)$, slope $\frac{1}{4}$ (e) $(-5, -5)$, slope 1 (f) $(0, b)$, slope m

4. Find the equation, in general form $Ax + By + C = 0$, of the line through the given point and with the given slope.
(a) $(5, 3)$, slope 2 (b) $(3, -6)$, slope $\frac{1}{5}$ (c) $(-2, 4)$, slope $-\frac{2}{3}$

(d) $(-7, -1)$, slope $\frac{3}{8}$ (e) $(0, -9)$, slope $-\frac{5}{6}$ (f) $(0, 0)$, slope $\frac{5}{2}$

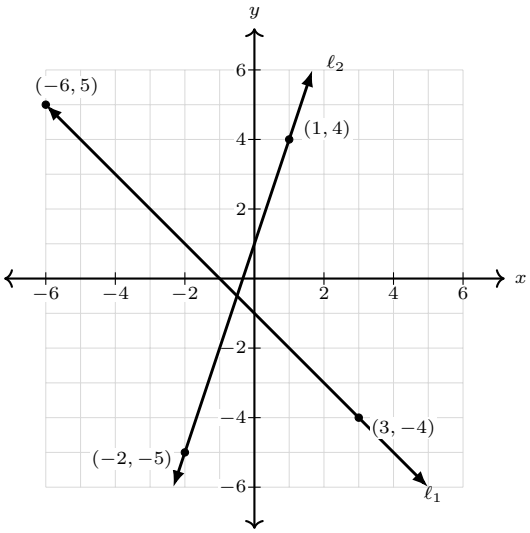
5. Each equation below is written in slope–point form. State the slope of the line and the coordinates of the point used to write the equation.

(a) $y - 4 = -\frac{7}{2}(x + 6)$ (b) $y + 8 = \frac{3}{4}x$ (c) $y - 1 = -5(x - 9)$

(d) $y = 6(x + 2)$ (e) $y + 2 = -\frac{4}{9}x$ (f) $y = \frac{7}{8}x$

6. Two lines have been drawn on the grid. Each line passes through at least two points with integer coordinates. Determine the equation of each line, in general form.

ℓ_1:



ℓ_2:

7. *(Multiple Choice)* The equation of the line passing through the point $(5, -3)$ with slope -4 is

A. $4x + y - 17 = 0$
B. $4x + y + 17 = 0$
C. $4x - y - 17 = 0$
D. $4x + y - 23 = 0$

8. *(Multiple Choice)* The equation of the line passing through the origin with slope $-\frac{3}{5}$ is

A. $3x + 5y = 0$
B. $3x - 5y = 0$
C. $5x + 3y = 0$
D. $5x - 3y = 0$

9. *(Multiple Choice)* Which of the following linear equations is equivalent to $y - 5 = -\frac{2}{7}(x + 3)$?

A. $2x - 7y + 29 = 0$
B. $2x + 7y - 29 = 0$
C. $2x + 7y + 29 = 0$
D. $2x - 7y - 29 = 0$

10. *(Numerical Response)* The equation of the line with an x-intercept of -4 and slope 9 can be written in general form $9x + By + C = 0$. The value of $B + C$ is _____.

Record your answer left to right. | | | | |

11. Find the equation, in general form, of the line through each pair of points.
(a) $(5, 3)$ and $(2, -3)$

(b) $(-4, 6)$ and $(2, -9)$

(c) $(-6, -11)$ and $(9, 19)$

(d) $(8, -13)$ and $(0, -9)$

12. Identify the lines in #11 which are
(i) parallel

(ii) perpendicular

13. Write the equation of each line in general form.
(a) with slope $-\frac{3}{5}$ and a y-intercept of 4

(b) with slope 5 and an x-intercept of -2

14. Write the equation of each line in general form.
(a) through the point $(6, 0)$ and perpendicular to $4x - 7y + 21 = 0$

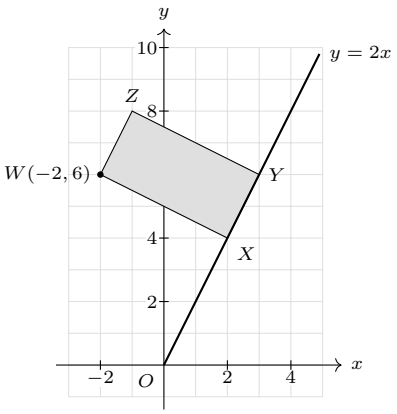
(b) through the point $(-2, 5)$ and parallel to $3x + 8y - 16 = 0$

15. Write the equation of each line in general form.
(a) perpendicular to $5x - 3y + 9 = 0$ and with the same y-intercept as $x + y - 8 = 0$

(b) with the same x-intercept as $7x - 4y + 28 = 0$, and through the point $(6, -8)$

16. Line l contains the point $A(5, 7)$ and is parallel to a line which contains the points $B(-3, 4)$ and $C(9, -2)$. Determine the equation of line l in the form $y = mx + b$.

17. A coordinate grid is placed over the blueprint of a campground. The ranger station sits at the origin, O, and rectangle $WXYZ$ represents a picnic pavilion. A straight nature trail, with equation $y = 2x$, runs from the station along side XY of the pavilion. Corner W is at $(-2, 6)$.



(a) Determine the equation of side WZ.

(b) Determine the equation of side WX.

(c) Determine the coordinates of corner X, and verify that X lies on the trail.

18. A customer with a fixed amount of store credit can buy 3 notebooks and 4 pens, or 5 notebooks and 1 pen. A linear relationship exists between the number of notebooks, x, and the number of pens, y, that can be bought.

(a) Write the coordinates of two points which lie on the graph of this linear relationship.

(b) Determine the equation of the linear relationship, in general form.

19. *(Multiple Choice)* The equation of the line through the point $(4, -9)$ and perpendicular to the line with equation $3x - 2y + 11 = 0$ can be written in the form

A. $y - 9 = \frac{2}{3}(x - 4)$ B. $y = -\frac{2}{3}(x + 4)$

C. $y + 9 = -\frac{3}{2}(x - 4)$ D. $y + 9 = -\frac{2}{3}(x - 4)$

20. *(Multiple Choice)* A line passing through the point $(0, -5)$ is perpendicular to the line $x - 3y - 6 = 0$. The equation of the line is

A. $3x + y + 5 = 0$ B. $3x + y - 5 = 0$

C. $x - 3y - 15 = 0$ D. $3x - y + 5 = 0$

21. *(Multiple Choice)* Which of the following linear relations is *not* equivalent to the other three?

A. $y - 5 = -\frac{1}{2}(x + 4)$ B. $x + 2y - 6 = 0$

C. the line passing through $(0, 3)$ and $(6, 0)$

D. $y = -\frac{1}{2}x + 2$

22. *(Multiple Choice)* The point $(3, 4)$ lies on a line with slope -2. The y-intercept of the line is

A. -10 B. -2 C. 2 D. 10

23. *(Numerical Response)* The line through the points $(-5, 6)$ and $(-2, -3)$ has equation $y + ax + b = 0$, where a and b are integers. The value of $a + b$ is _____.

Record your answer left to right. | | | | |

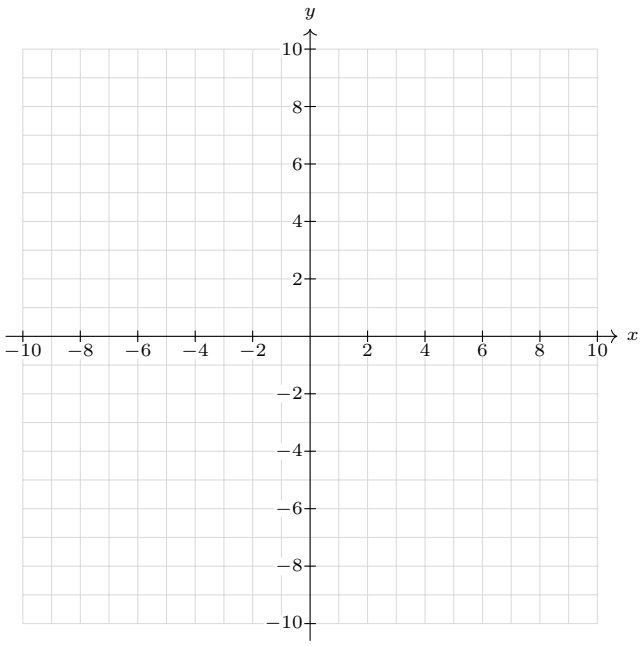
24. Ana and Ben each wrote an equation for the line through $(1, 3)$ and $(-3, -5)$. Ana wrote $y - 3 = 2(x - 1)$ and Ben wrote $y + 5 = 2(x + 3)$.

(a) Explain why both equations are correct.

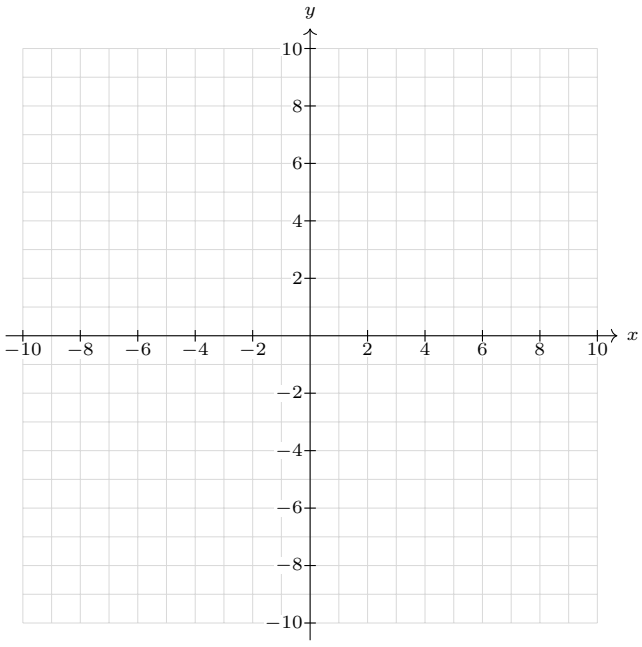
(b) Verify that both equations simplify to the same equation in slope–intercept form.

Lesson 5 Assignment *Show your work.*

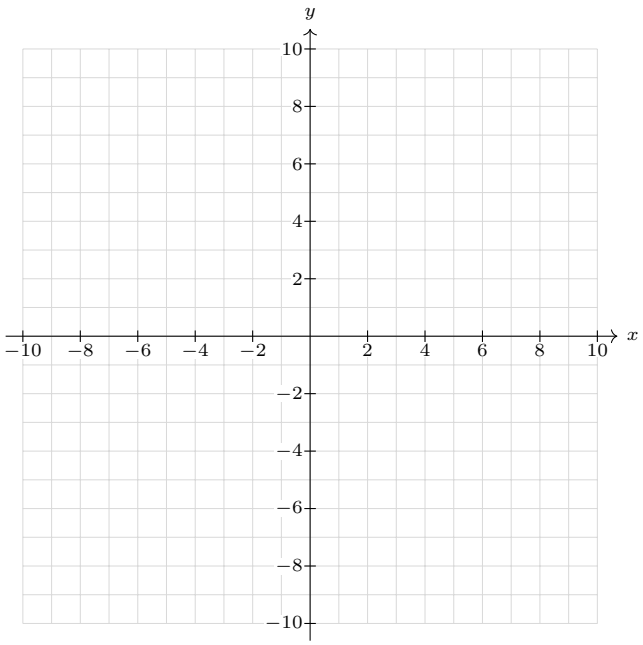
1. Without altering the form in which the equation is written, explain how to graph $y = -2x + 8$. Draw the graph on the grid provided.



2. Without altering the form in which the equation is written, explain how to graph $3x - 4y - 24 = 0$. Draw the graph on the grid provided.

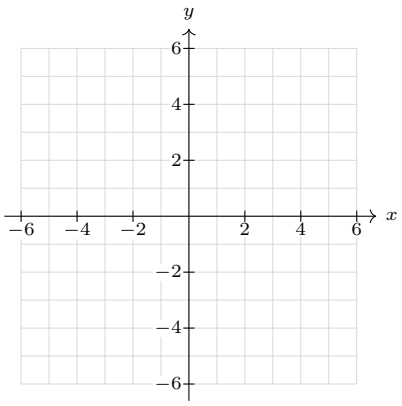


3. Without altering the form in which the equation is written, explain how to graph $y - 3 = -3(x + 2)$. Draw the graph on the grid provided.

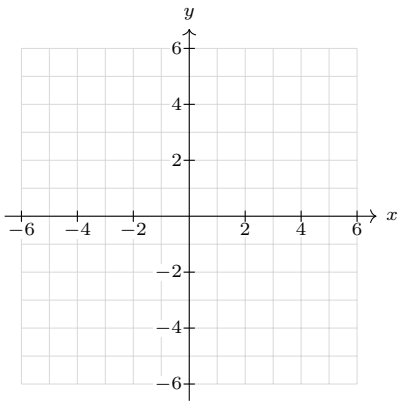


4. Without altering the form in which each equation is written, draw the graph of each of the following on the grid provided. Below each grid, state the strategy you used (the starting point and the slope, the two intercepts, or the points you found).

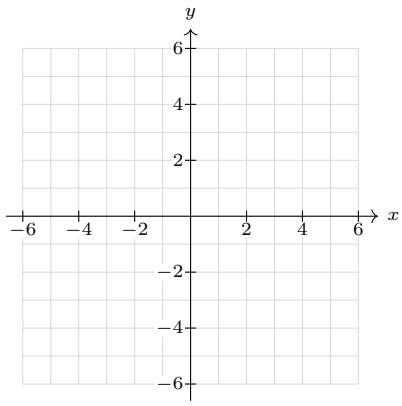
(a) $y = -\frac{1}{2}x + 4$



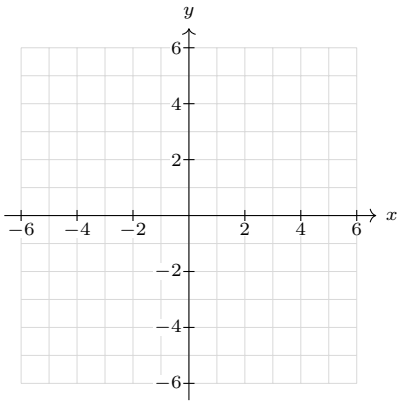
(b) $y - 2 = -3(x + 4)$



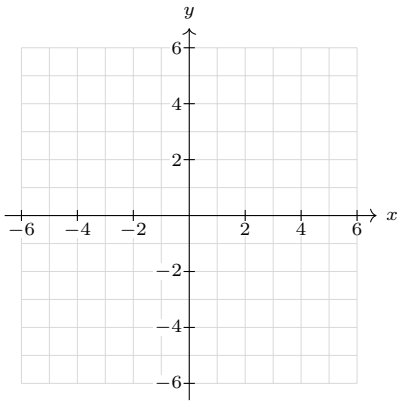
(c) $2x - 5y - 10 = 0$



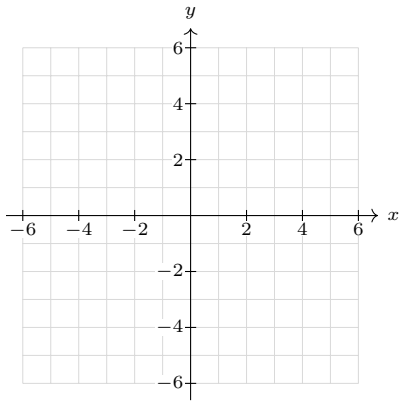
(d) $y - 2 = -\frac{3}{4}(x + 4)$



(e) $2x + 5y = 0$



(f) $3x + 2y - 3 = 0$

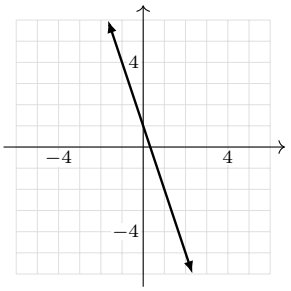


5. Match each linear relation to its graph below. Each graph is used exactly once.

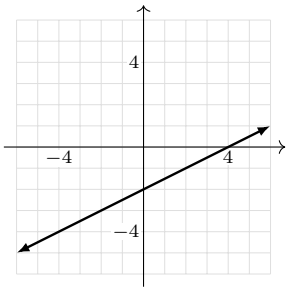
Equation 1: $y + 1 = 2(x - 1)$ **Equation 2:** $x - 2y - 4 = 0$

Equation 3: $y = -3x + 1$ **Equation 4:** $x + 2y - 8 = 0$

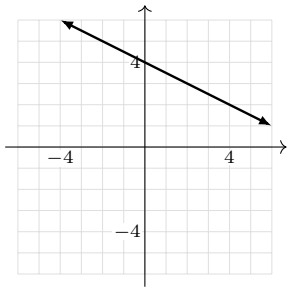
Graph A



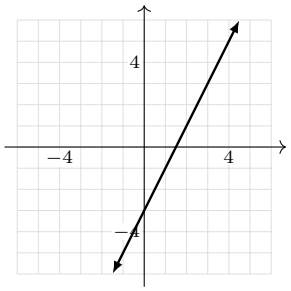
Graph B



Graph C



Graph D



Eqn 1 → Graph _____ Eqn 2 → Graph _____ Eqn 3 → Graph _____ Eqn 4 → Graph _____

Show the slope and y-intercept you used for each equation.

6. **Sorting equivalent forms.** The eight equations below describe only a few different lines. Sort them into groups, where every equation in a group describes the same line. Justify each group by rewriting every equation in slope–intercept form.

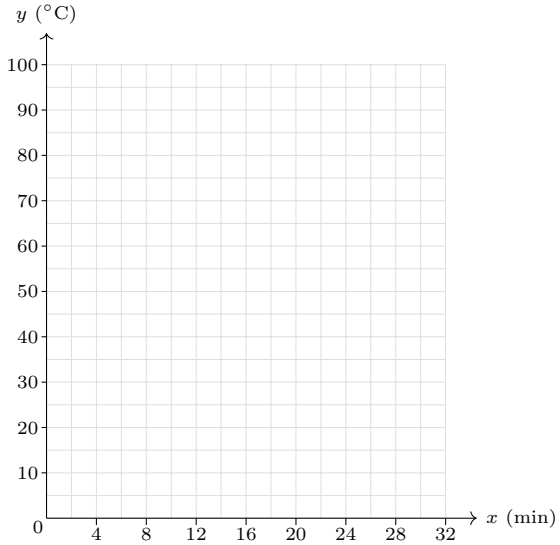
A. $y = -\frac{2}{3}x + 2$ **B.** $2x - 3y - 6 = 0$ **C.** $y - 4 = -\frac{2}{3}(x + 3)$ **D.** $2x + 3y + 6 = 0$

E. $y = \frac{2}{3}x - 2$ **F.** $y + 2 = -\frac{2}{3}(x - 6)$ **G.** $2x + 3y - 6 = 0$ **H.** $y - 2 = \frac{2}{3}(x - 6)$

7. The data below is taken from a continuous linear relationship between the number of minutes, x, since a cup of coffee was poured and the temperature, y, of the coffee in degrees Celsius.

x (min)	0	2	4	6	8	10	12	14
y (°C)	92	86	80	74	68	62	56	50

(a) Plot the data on the grid and determine, in general form, the equation of the linear relation the data represents.



(b) Rewrite the equation of the linear relation (i) in terms of x, and (ii) in terms of y.

(c) What real-world quantity does the slope represent? What does the y-intercept represent?

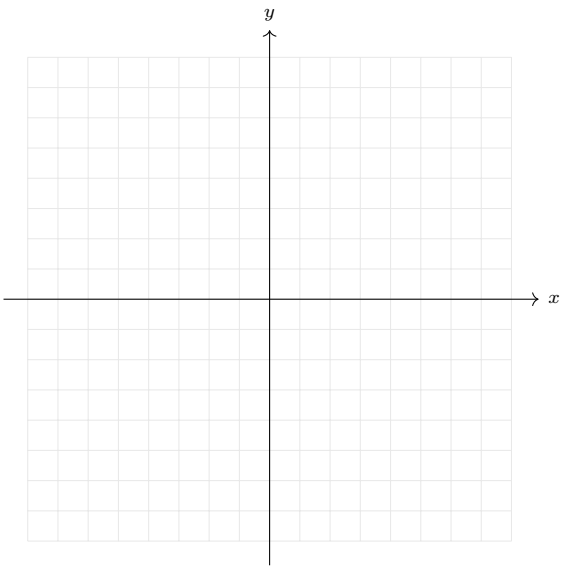
(d) Real coffee cools more and more slowly and levels off at room temperature (about 20°C). What does this linear model predict for the temperature at $x = 30$ min? For what domain is the linear model reasonable?

8. Consider the linear relations

$$x + 4y + 8 = 0 \qquad \text{and} \qquad y - 2 = 2(x + 3).$$

(a) Determine the x- and y-intercepts of *each* line.

(b) Graph both lines on the axes below. Choose a scale for each axis so that all four intercepts are visible, and label the scale you chose.



9. *(Multiple Choice)* Without altering its form, which strategy correctly describes how to graph $y - 5 = -4(x + 1)$?
A. Plot $(0, 5)$; use slope -4.
B. Plot $(-1, 5)$; use slope -4.
C. Plot $(-1, 5)$; use slope 4.
D. Plot $(1, -5)$; use slope -4.

10. *(Multiple Choice)* A line has general form $5x - 2y + 20 = 0$. Its x-intercept is
A. -4 B. 4 C. -10 D. 10

11. *(Numerical Response)* The equation $y - 7 = 3(x + 2)$ is rewritten in general form $ax - y + c = 0$, with $a > 0$. The value of $a + c$ is _____.

Record your answer left to right. | | | | |

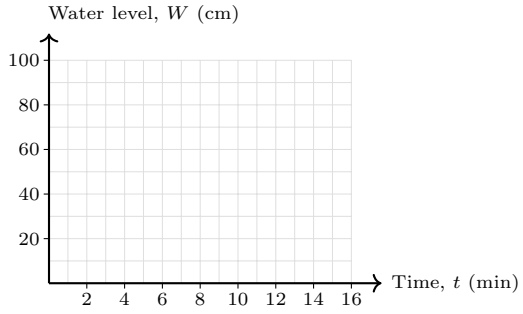
Lesson 6 Assignment *Show your work.*

1. GreenTech Solar was founded in 2011. By March 2013 the company had sold 340 000 solar panels, and by September 2013 it had sold 460 000 panels. Calculate the average rate of change, stating appropriate units.

2. In 2016 a city's bike-share program recorded 8 142 900 rides. In 2011 the program recorded 11 398 650 rides. Calculate the average rate of change, stating appropriate units.

3. Water is draining from a cylindrical tank at a constant rate. After 5 minutes the water level is 60 cm; after 10 minutes the water level is 30 cm.

(a) Plot the ordered pairs (time, water level) on the grid and join them.



(b) Calculate the slope of the line segment joining the ordered pairs.

(c) Explain what the slope of the graph represents.

(d) Complete: the water level is changing at the rate of _____ ____ per _____.

(e) Determine an equation in the form $W = mt + b$, where W is the water level in cm and t is the time in minutes.

(f) After how many minutes will the tank be empty?

(g) What was the water level in the tank when it started draining?

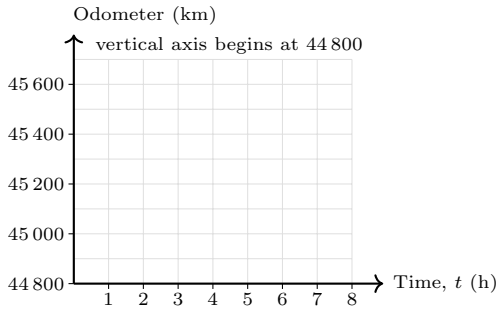
(h) The tank is cylindrical with a radius of 10 cm, so $V = \pi r^2 h = 100\pi h$.

i) Calculate the volume (in terms of π) of water in the tank after 5 min and after 10 min.

ii) Calculate the rate (in terms of π) at which water is draining out of the tank.

4. Amara rented a car. After driving for 3 hours the odometer reading was 45 120, and after 7 hours the odometer reading was 45 516. Her journey was completed after 8 hours.

(a) Plot ordered pairs (time, odometer reading) on the grid and join them.



(b) Calculate the slope of the line segment joining the ordered pairs.

(c) Explain what the slope of the graph represents.

(d) Assuming a constant rate of driving for the whole journey, determine the odometer reading at the start of the journey.

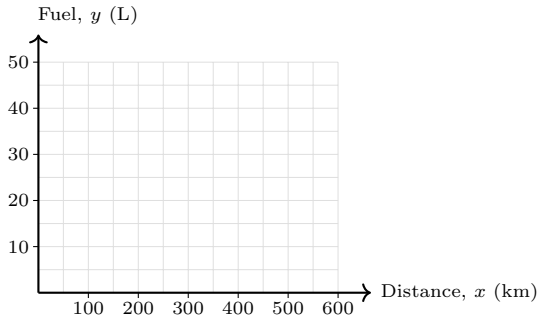
(e) Determine an equation in the form $f(t) = mt + b$, where $f(t)$ is the odometer reading after t hours.

(f) State an appropriate domain and range for f.

(g) Determine $f(5)$ and explain what it means. Then determine t when $f(t) = 45\,417$.

5. To test the fuel consumption of a new hybrid car, Devon filled the gas tank with 45 L and drove until it was empty, covering a total distance of 540 km.

(a) Sketch a graph with distance travelled on the horizontal axis and fuel remaining in the tank on the vertical axis.



(b) Write an equation in the form $y = mx + b$ representing the volume of fuel in the tank as a function of distance.

(c) State the slope of the graph, and explain what it represents.

(d) Determine the distance travelled when 15 L of gas has been used.

(e) How many litres of gas are used by the car when it has travelled 300 km?

6. After 5 days of heavy rain the water level in a river peaked at 3.60 m above its regular level. Six days later the water level had dropped to 2.40 m above its regular level.

(a) Assuming the water level falls at a constant rate, determine a function $h(t)$ giving the height of the river above its regular level as a function of time. Take $t = 0$ at the peak.

(b) State the slope of the graph of the function, and explain what it represents.

(c) Determine an appropriate domain and range for h.

7. *(Multiple Choice)* A plumbing company charges a fixed call-out fee plus a fixed hourly rate for any repair job. A 2-hour job costs \$130 and a 5-hour job costs \$250. The fixed call-out fee and the cost of a 6-hour job are, respectively,

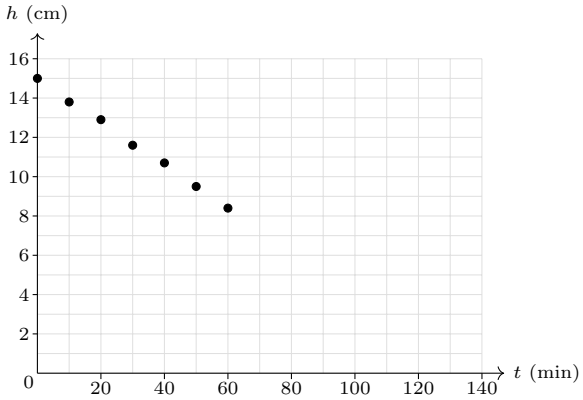
A. \$40 and \$280
B. \$50 and \$290
C. \$40 and \$290
D. \$50 and \$280

8. *(Numerical Response)* The temperature at the top of a mine shaft is 14°C. At 300 metres below the surface the temperature is 15.02°C. To the nearest tenth, the rate of temperature increase in °C per km is _____.

Record your answer left to right. | | | | |

9. *(Extension)* **Real data are rarely perfectly linear.** A class lit a candle and measured its height, h centimetres, every 10 minutes.

Time, t (min)	0	10	20	30	40	50	60
Height, h (cm)	15.0	13.8	12.9	11.6	10.7	9.5	8.4



(a) The data are plotted on the grid. Use a ruler to draw the straight line that you think best fits the points.

(b) Choose two points on *your line* (not necessarily data points) and write the equation of your line in the form $h = mt + b$. What does the slope mean?

(c) Use your equation to predict when the candle will burn out.

Lesson 7 Assignment *Show your work.*

1. *(Multiple Choice)* Which of these ordered pairs can be found on the graph of the line $4x - 6y - 9 = 0$?
i) $(6, 2.5)$ *ii*) $(-4, -3.9)$ *iii*) $(0, -1.5)$ *iv*) $(-3, 3)$
A. *i*) and *ii*) only B. *i*) and *iii*) only
C. *i*), *ii*), and *iii*) only D. some other combination

2. *(Multiple Choice)* If the lines $ax + by + c = 0$ and $mx + ny + p = 0$ are parallel, then
A. $an - bm = 0$ B. $an + bm = 0$
C. $am - bn = 0$ D. $am + bn = 0$

3. *(Numerical Response)* The line through $(11, -2)$ that is perpendicular to the line $2x - 3y - 7 = 0$ also passes through the point $(p, 4)$. The value of p is _____.
Record your answer left to right. ☐☐☐☐

4. *(Multiple Choice)* The line passing through the points $(-6, -3)$ and $(-3, -1)$ has equation
A. $2x - 3y + 3 = 0$ B. $2x - 3y - 3 = 0$
C. $2x + 3y + 3 = 0$ D. $2x + 3y - 3 = 0$

5. *(Numerical Response)* The lines $5x - y + 8 = 0$ and $3x - ky + 40 = 0$, where $k \in W$, intersect on the y-axis. The value of k is _____.
Record your answer left to right. ☐☐☐☐

6. *(Multiple Choice)* Line ℓ_1 has equation $y = 3x + 2$, and line ℓ_2 has x-intercept 4. Which equation represents a line perpendicular to ℓ_1 with the same x-intercept as ℓ_2?
A. $x + 3y - 4 = 0$ B. $x + 3y + 4 = 0$
C. $3x + y - 12 = 0$ D. $3x + y + 12 = 0$

7. *(Numerical Response)* The equation of a certain line is $Ax + 5y + C = 0$. The line has slope $-\frac{7}{5}$ and y-intercept -4. The value of $\frac{A}{C}$, to the nearest hundredth, is _____.
Record your answer left to right. ☐☐☐☐

8. *(Multiple Choice)* The line ℓ_1 passes through $(-4, 3)$ and $(-2, -9)$. Which statements are true?
i) ℓ_1 passes through $(2, -33)$. *ii*) ℓ_1 has an x-intercept of $-\frac{7}{2}$.
iii) ℓ_1 is perpendicular to $y = \frac{1}{6}x + 5$.
A. *i*) and *ii*) only B. *i*) and *iii*) only
C. *ii*) and *iii*) only D. *i*), *ii*), and *iii*)

9. *(Written Response)* Consider the points $P(-5,-3)$, $Q(3,3)$, $R(-6,4)$ and $S(9,-2)$.

(a) Show that the equation of the line L_1, through S and perpendicular to PQ, is $y = -\frac{4}{3}x + 10$.

(b) Determine the equation of the line L_2, through R and parallel to PQ. Give the answer in slope–intercept form.

(c) Determine the x- and y-intercepts of each line. Then sketch L_1 and L_2 on one set of axes, choosing a scale on each axis so that all four intercepts are visible.

10. State the equation of each line in slope–point form, $y-y_1 = m(x-x_1)$, and then rewrite it in general form, $Ax+By+C = 0$.

(a) slope -2, through $(3,-7)$

(b) slope $\frac{3}{4}$, through $(-8,5)$

(c) through $(-2,-6)$ and $(4,3)$; then show that both points satisfy your general-form equation

11. Consider the points $A(-4,7)$, $B(-4,-2)$ and $C(3,7)$.

(a) Determine the equation of the line through A and B.

(b) Determine the equation of the line through A and C.

(c) State the slope of each of the two lines.

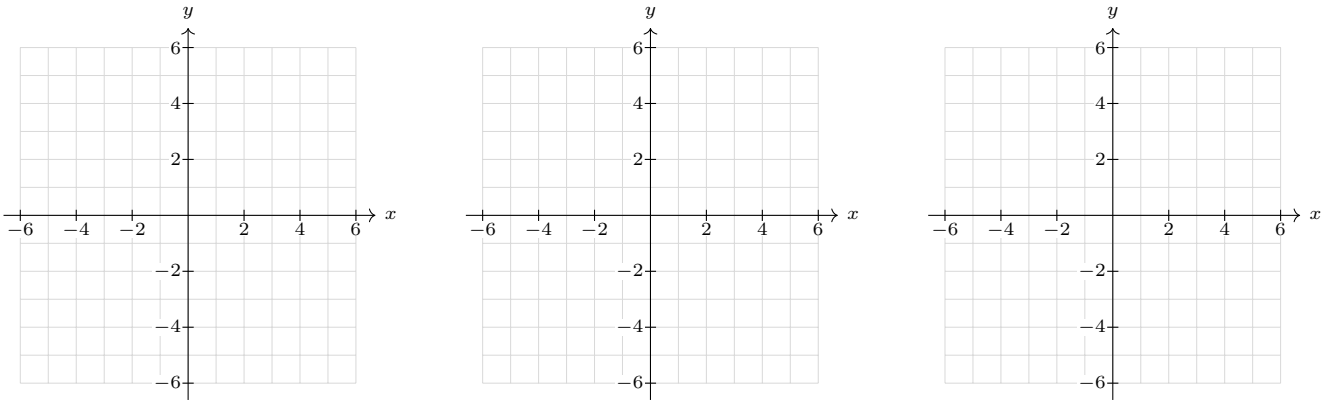
(d) Explain why one of these two lines cannot be written in the form $y = mx + b$.

(e) Determine the equation of the line through B that is parallel to line AC.

(f) Determine the equation of the line through C that is perpendicular to line AC.

12. Graph each line on the grid provided, using the method best suited to the form the equation is given in.

(a) $y = -\frac{2}{3}x + 4$ (b) $3x - 4y - 12 = 0$ (c) $x = -5$



13. A hot-air balloon descends at a constant rate. Its height h, in metres, is recorded against the time t, in minutes, since the descent began.

Time, t (min)	0	4	8	12
Height, h (m)	1 450	1 310	1 170	1 030

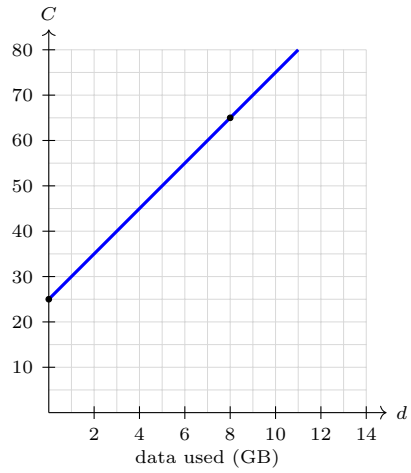
(a) Determine the rate of change of height with respect to time. Include units, and state what it means in this context.

(b) Write an equation for h in terms of t.

(c) Determine the height of the balloon 25 minutes into the descent.

(d) Determine, to the nearest tenth of a minute, when the balloon reaches the ground.

14. The graph shows the monthly cost C, in dollars, of a cell phone plan against the amount of data used, d, in gigabytes.



(a) Determine the slope of the graph. Include units, and interpret it in context.

(b) Interpret the vertical intercept in context.

(c) Write an equation for C in terms of d.

(d) Determine the cost of a month in which 13.5 GB of data is used.

15. *(Written Response)* Marisol is paid a fixed weekly base amount plus a commission that is a fixed percentage of her sales. In a week with \$4 000 in sales she was paid \$760; in a week with \$9 500 in sales she was paid \$1 035.

(a) Determine her commission rate, expressed as a percentage of sales.

(b) Determine her weekly base amount.

(c) Write an equation for her weekly pay P in terms of her sales s.

(d) Determine the sales she needs in order to be paid \$1 500 in a week.

Answer Key — Lesson Assignments

Lesson 1 Assignment

1. LINEAR: (a), (c), (f), (h). NON-LINEAR: (b), (d), (e), (g). **2.** (a) $m = 9$, $b = -1$ (b) $m = \frac{5}{8}$, $b = 4$ (c) $m = -\frac{2}{3}$, $b = 10$ (d) $m = 3$, $b = -4$ (e) $m = p$, $b = q$. **3.** (a) $y = 6x - 7$ (b) $y = \frac{1}{3}x + \frac{3}{4}$ (c) $y = -5x$ (d) $y = kx + n$. **4.** (a) $m = \frac{1}{2}$, $b = -3$ (b) $m = -1$, $b = 4$ (c) $m = -\frac{5}{6}$, $b = 0$ (d) $m = 0$, $b = -2$ (horizontal line). **5.** (a) y-int $= 8$, x-int $= -2$ (b) y-int $= -6$, x-int $= -6$ (c) y-int $= -8$, x-int $= 6$ (d) y-int $= 3$, x-int $= 4$. **6.** Question 4's method needs one point and the slope; $y = 8x$ has y-intercept 0, so $(0, 0)$ and $m = 8$ suffice. Question 5's method needs two *different* intercept points, but both intercepts of $y = 8x$ are $(0, 0)$. **7.** (a) $m = -1$, $b = 0$ (b) both intercepts are $(0, 0)$ (d) $D = \{x \mid x \in R\}$, $R = \{y \mid y \in R\}$ (e) mirror images of each other across the x-axis (equivalently across the y-axis). **8.** (i) $m = 9$, $b = -14$, x-int ≈ 1.56 (ii) $m = -\frac{29}{2} = -14.5$, $b = -40$, x-int ≈ -2.76 (iii) $m = -\frac{7}{4} = -1.75$, $b = 50$, x-int ≈ 28.57. **9.** C. **10.** D. **11.** C. **12.** D. **13.** 1.6. **14.** 8.9.

Lesson 2 Assignment

1. (a) $y = 6x - 2$ (b) $y = -\frac{3}{7}x + 5$ (c) $y = -\frac{4}{9}x$ (d) $y = x + 8$ (e) $y = \frac{6}{5}x - 11$ (f) $y = \frac{3}{8}x - 9$ (g) $y = -\frac{1}{7}x - 2$ (h) $y = \frac{1}{12}x + 10$ (i) $y = -\frac{2}{9}x + 11$ (j) $y = -\frac{1}{r}x + q$. **2.** n_1 : $y = \frac{1}{3}x - 4$; n_2 : $y = -3x + 6$; n_3 : $y = 2x + 1$. **3.** (a) x-int 8, y-int -4, $m = \frac{1}{2}$, $y = \frac{1}{2}x - 4$ (b) x-int -6, y-int 9, $m = \frac{3}{2}$, $y = \frac{3}{2}x + 9$. **4.** $y = -4x - 14$. **5.** $y = 3x + 6$. **6.** (a) $x = -8$ (b) $y = 5$ (c) $y = -6$ (d) $x = a$. **7.** (a) $m = 0$, $b = 5$ (b) no x-intercept, y-intercept 5 (c) domain $\{x \mid x \in R\}$, range $\{y \mid y = 5\}$ (d) $(3, 5)$. **8.** (a) x-intercept -7, no y-intercept (b) fails the vertical line test — the one input $x = -7$ has infinitely many outputs (c) domain $\{x \mid x = -7\}$, range $\{y \mid y \in R\}$ (d) $x = 3$. **9.** (a) $y = -2$ (b) $x = 7$ (c) $x = -3$ (d) $y = 9$ (e) $y = 0$ (f) $x = 0$ (g) $x = 2$ (h) $x = -4$ (i) $y = 7$. **10.** A. **11.** D. **12.** D. **13.** 8. **14.** 1.8.

Lesson 3 Assignment

1. (a) $6x - y - 5 = 0$ (b) $3x + y - 8 = 0$ (c) $px - y + q = 0$ (d) $5x + 6y - 18 = 0$ (e) $6x - 8y + 1 = 0$ (f) $21x - 6y - 2 = 0$. **2.** (a) $m = -1$, y-int $= 15$ (b) $m = \frac{4}{3}$, y-int $= 8$ (c) $m = \frac{9}{4}$, y-int $= -2$ (d) $m = -\frac{1}{2}$, y-int $= \frac{3}{2}$ (e) $m = \frac{1}{2}$, y-int $= 4$ (f) $m = -\frac{5}{4}$, y-int $= 5$. **3.** (a) $m = -3$, y-int $= 9$, x-int $= 3$ (b) $m = \frac{7}{4}$, y-int $= 7$, x-int $= -4$ (c) $m = \frac{6}{7}$, y-int $= -\frac{4}{7}$, x-int $= \frac{2}{3}$. **4.** $5x - 2y - 8 = 0$. **5.** $5x + 4y - 96 = 0$. **6.** (a) $k = -12$ (b) $k = 6$. **7.** (a) $a = -2$ (b) $a = \frac{3}{2}$. **8.** (i) E (ii) C (iii) A (iv) F (v) G. **9.** A. **10.** C. **11.** D. **12.** A. **13.** D. **14.** C. **15.** C. **16.** 5.

Lesson 4 Assignment

1. (a) $y - 2 = 3(x - 7)$ (b) $y + 4 = -2(x - 5)$ (c) $y - 8 = 5(x + 6)$ (d) $y + 5 = \frac{1}{3}x$ (e) $y = \frac{3}{5}(x + 10)$ (f) $y + \frac{3}{2} = \frac{4}{7}(x + 2)$. **2.** (a) $y = 6x - 14$ (b) $y = -5x + 3$ (c) $y = -9x - 39$ (d) $y = 3x + 37$. **3.** (a) $y = 4x - 5$ (b) $y = 5x - 23$ (c) $y = -3x + 6$ (d) $y = \frac{1}{4}x + 5$ (e) $y = x$ (f) $y = mx + b$. **4.** (a) $2x - y - 7 = 0$ (b) $x - 5y - 33 = 0$ (c) $2x + 3y - 8 = 0$ (d) $3x - 8y + 13 = 0$ (e) $5x + 6y + 54 = 0$ (f) $5x - 2y = 0$. **5.** (a) $m = -\frac{7}{2}$, $P(-6, 4)$ (b) $m = \frac{3}{4}$, $P(0, -8)$ (c) $m = -5$, $P(9, 1)$ (d) $m = 6$, $P(-2, 0)$ (e) $m = -\frac{4}{9}$, $P(0, -2)$ (f) $m = \frac{7}{8}$, $P(0, 0)$. **6.** $\ell_1 \Rightarrow x + y + 1 = 0$; $\ell_2 \Rightarrow 3x - y + 1 = 0$. **7.** A. **8.** A. **9.** B. **10.** 35. **11.** (a) $2x - y - 7 = 0$ (b) $5x + 2y + 8 = 0$ (c) $2x - y + 1 = 0$ (d) $x + 2y + 18 = 0$. **12.** (i) a and c (ii) a and d, and c and d. **13.** (a) $3x + 5y - 20 = 0$ (b) $5x - y + 10 = 0$. **14.** (a) $7x + 4y - 42 = 0$ (b) $3x + 8y - 34 = 0$. **15.** (a) $3x + 5y - 40 = 0$ (b) $4x + 5y + 16 = 0$. **16.** $y = -\frac{1}{2}x + \frac{19}{2}$. **17.** (a) $y = 2x + 10$ (b) $y = -\frac{1}{2}x + 5$ (c) $X(2, 4)$; $2(2) = 4$, so X is on the trail. **18.** (a) $(3, 4)$ and $(5, 1)$ (b) $3x + 2y - 17 = 0$. **19.** D. **20.** A. **21.** D. **22.** D. **23.** 12. **24.** (a) Both use the correct slope, 2, and a point that is on the line; slope–point form works with any point on the line. (b) Both simplify to $y = 2x + 1$.

Lesson 5 Assignment

1. Plot the y-intercept $(0, 8)$; with slope $-2 = \frac{-2}{1}$, step down 2 and right 1 to $(1, 6)$, repeat, and join the points (x-intercept $(4, 0)$). **2.** x-intercept: $y = 0 \Rightarrow x = 8$; y-intercept: $x = 0 \Rightarrow y = -6$. Plot $(8, 0)$ and $(0, -6)$ and draw the line through them. **3.** Plot the point $(-2, 3)$; with slope -3, step down 3 and right 1 to $(-1, 0)$, repeat, and join the points. **4.** (a) start at $(0, 4)$, slope $-\frac{1}{2}$ (b) start at $(-4, 2)$, slope -3 (c) intercepts $(5, 0)$ and $(0, -2)$ (d) start at $(-4, 2)$, slope $-\frac{3}{4}$: $(0, -1)$, $(4, -4)$ (e) $(0, 0)$ and another point such as $(5, -2)$ (f) $(1, 0)$, $(3, -3)$, $(-1, 3)$ (the y-intercept $\frac{3}{2}$ is a fraction). **5.** 1D, 2B, 3A, 4C. **6.** A, C, F, G: $y = -\frac{2}{3}x + 2$; B, E, H: $y = \frac{2}{3}x - 2$; D alone: $y = -\frac{2}{3}x - 2$ (parallel to the first group). **7.** (a) $3x + y - 92 = 0$ (b)(i) $y = -3x + 92$ (ii) $x = \frac{92 - y}{3}$ (c) the slope -3 is the cooling rate, 3°C lost per minute; the y-intercept 92 is the coffee's temperature, in °C, at the moment it was poured (d) 2°C at 30 min — impossible, since coffee cannot cool below room temperature; the model is reasonable only for about the measured times, $\{x \mid 0 \le x \le 14,\ x \in R\}$. **8.** (a) $x + 4y + 8 = 0$: x-int -8, y-int -2; $y - 2 = 2(x + 3)$: x-int -4, y-int 8 (b) e.g. count x by 2 and y by 1, so the grid shows $-16 \le x \le 16$ and $-8 \le y \le 8$ (any scale showing the four intercepts); each line drawn through its two intercepts. **9.** B. **10.** A. **11.** 16.

Lesson 6 Assignment

1. 20 000 solar panels per month. **2.** $-651\,150$ rides per year (ridership is falling). **3.** (b) -6 cm/min (c) the rate at which the water level is changing, in cm/min (d) decreasing at 6 cm per minute (e) $W = -6t + 90$ (f) 15 min (g) 90 cm (h) i) 6000π cm^3 and 3000π cm^3 ii) 600π cm^3/min draining out. **4.** (b) 99 km/h (c) the average speed of the car, in km/h (d) 44 823 (e) $f(t) = 99t + 44\,823$ (f) domain $\{t \mid 0 \le t \le 8,\ t \in R\}$; range $\{f(t) \mid 44\,823 \le f(t) \le 45\,615,\ f(t) \in R\}$ (g) $f(5) = 45\,318$: the odometer read 45 318 km after 5 h; $t = 6$ h. **5.** (b) $y = -\frac{1}{12}x + 45$ (c) $m = -\frac{1}{12}$; the fuel is decreasing at $\frac{1}{12}$ L per km (d) 180 km (e) 25 L. **6.** (a) $h(t) = -0.2t + 3.60$ (b) $m = -0.2$; the level is falling at 0.2 m per day (c) domain $\{t \mid 0 \le t \le 18,\ t \in R\}$; range $\{h(t) \mid 0 \le h(t) \le 3.60,\ h(t) \in R\}$. **7.** B. **8.** 3.4. **9.** *(Extension)* Answers vary. (b) e.g. through $(0, 15)$ and $(60, 8.4)$: $h = -0.11t + 15$; the candle shrinks about 0.11 cm per minute (c) about 136 min.

Lesson 7 Assignment

1. B. **2.** A. **3.** 7. **4.** A. **5.** 5. **6.** A. **7.** 0.35 (record .35). **8.** D. **9.** (a) $y = -\frac{4}{3}x + 10$ (b) $y = \frac{3}{4}x + 8.5$ (c) L_1: $(7.5, 0)$, $(0, 10)$; L_2: $\left(-\frac{34}{3}, 0\right) \approx (-11.3, 0)$, $(0, 8.5)$; e.g. axes $-12 \le x \le 8$, $-2 \le y \le 12$, counting by 2. **10.** (a) $y + 7 = -2(x - 3)$; $2x + y + 1 = 0$ (b) $y - 5 = \frac{3}{4}(x + 8)$; $3x - 4y + 44 = 0$ (c) $y - 3 = \frac{3}{2}(x - 4)$; $3x - 2y - 6 = 0$; check: $3(-2) - 2(-6) - 6 = 0$ and $3(4) - 2(3) - 6 = 0$. **11.** (a) $x = -4$ (b) $y = 7$ (c) AB: undefined; AC: 0 (d) AB — an undefined slope gives no value of m to substitute (e) $y = -2$ (f) $x = 3$. **12.** (a) through $(0, 4)$ and $(3, 2)$ (b) through $(4, 0)$ and $(0, -3)$ (c) vertical line through $(-5, 0)$. **13.** (a) -35 m/min; the balloon loses 35 m of height each minute (b) $h = -35t + 1\,450$ (c) 575 m (d) 41.4 min. **14.** (a) \$5/GB; each additional GB adds \$5 to the monthly cost (b) \$25 is the fixed monthly charge when no data is used (c) $C = 5d + 25$ (d) \$92.50. **15.** (a) 5% (b) \$560 (c) $P = 0.05s + 560$ (d) \$18 800.